

T-DDA: A Dynamic Programming-Based Framework for Team-oriented Dynamic Difficulty Adjustment in Collaborative Environments

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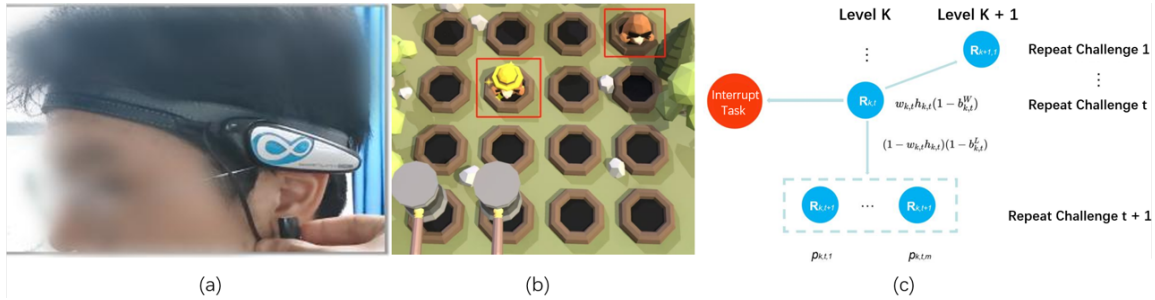


Figure 1: (a)The actual wearing effect of BrainLink Pro (b)Special and ordinary moles,the one wearing a yellow hat is a special gopher, and the other is a regular gopher (c)Probability model for player progression transitions for multiple rounds .

ABSTRACT

In interactive tasks involving team collaboration, enhancing team flow experience contributes to optimal team performance. To achieve this goal, we propose a Dynamic Programming-based framework for team-oriented Dynamic Difficulty Adjustment (T-DDA). T-DDA optimizes task difficulty to maximize engagement and induce team flow. Using a Markov-based progression model, the framework adaptively adjusts challenge levels based on real-time performance and BCI-monitored concentration. Preliminary findings from a pilot study (N=12) indicate that T-DDA holds promise for enhancing team cohesion relative to non-DDA systems. By shifting to team-based optimization, T-DDA provides a robust mathematical approach to sustaining group flow in multi-user virtual environments.

Index Terms: Group flow, multiplayer virtual reality, dynamic difficulty adjustment, dynamic programming.

1 INTRODUCTION

With the precipitous advancement of multi-user Virtual Reality (VR) technology, applications such as telemedical collaboration, emergency response simulations, and collaborative industrial assembly have emerged as predominant research and industrial frontiers[2, 3]. Within these complex collaborative operations, the primary objective for augmenting collective performance lies in the induction of "group flow"[4, 5]. This state is characterized as a pinnacle psychological condition where team members, aligned by a shared objective, achieve profound immersion, fluid kinetic coordination, and seamless synergy. Teams attaining group flow exhibit not only superior task efficiency but also a significant reduction in inter-member psychological distance, thereby fundamentally optimizing the collaborative experience.

In multi-agent collaborative tasks, inherent disparities in mem-

bers' expertise and skill levels render traditional static difficulty settings susceptible to a severe "coordination gap." [1] This is primarily manifested in the coexistence of anxiety and boredom: excessive task demands plunge low-skill members into anxiety, whereas insufficient challenges result in boredom for high-skill members. Such psychological asymmetry disrupts the collective rhythm, precipitating a breakdown of collaborative mechanisms and precluding the maintenance of sustained psychological immersion. While Dynamic Difficulty Adjustment (DDA) is extensively deployed in single-player contexts, its direct transposition to multi-user environments reveals substantial limitations. Prevailing algorithms are predominantly optimized for individual agents, thereby overlooking the critical dimension of inter-member interdependency. Furthermore, conventional approaches typically emphasize instantaneous difficulty balancing, lacking a global, long-term optimization strategy for team engagement across the entire training lifecycle.

To address these challenges, this study proposes a novel Team-oriented Dynamic Difficulty Adjustment (T-DDA) framework. Unlike conventional DDA[6] which relies on reactive, local heuristics, T-DDA employs Dynamic Programming to achieve global optimization across the entire collaboration session.

Stochastic Modeling via Markov Processes: The collaborative training sequence is modeled as a series of stochastic state transitions, precisely quantifying the probabilistic logic underlying "progression, repetition, and interruption."

Dynamic Programming (DP) Optimization: By implementing DP algorithms, the framework transcends local balancing to pursue the maximization of "global expected engagement" throughout the task duration.

Multi-modal Sensing and Feedback: By integrating EEG-based concentration metrics with real-time performance, we establish a closed-loop monitoring system for team states.

2 MATHEMATICAL MODELING AND OPTIMIZATION OF T-DDA

To formalize team-oriented adjustment, the training process is modeled as a discrete-time Markovian progression model. The state

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space $s_{k,t}$ denotes difficulty level k at attempt t . Transitions include: **Progression** ($s_{k,t} \rightarrow s_{k+1,1}$): success occurs when performance meets thresholds with high cognitive synergy (Group Flow); **Repetition** ($s_{k,t} \rightarrow s_{k,t+1}$): the system initiates sub-tasks at the same level to rebuild rapport if synergy is suboptimal; and **Interruption** (s_{break}): a terminal state caused by disengagement due to an imbalanced challenge-skill ratio.

Unlike traditional DDA seeking local stability, T-DDA maximizes the Expected Total Engagement ($R_{1,1}$), ensuring long-term immersion. The recursive objective function is formulated as:

$$R_{k,t} = w_{k,t} h_{k,t} (1 - b_{k,t}^W) R_{k+1,1} + (1 - w_{k,t} h_{k,t}) (1 - b_{k,t}^L) R_{k,t+1} + 1 \quad (1)$$

where:

- $w_{k,t} \in [0, 1]$: Probability of team goal attainment.
- $h_{k,t}$: Composite score integrating kinetic performance and BCI-sensed flow status.
- b^W, b^L : Probabilities of task abandonment after success or failure, respectively.

The probability model for player progression transitions is shown in Fig1(c). The optimal difficulty trajectory is derived by solving the Bellman Equation through backward induction. This approach allows the system to be strategically proactive—optimizing the difficulty path to guide the team toward a sustained "Flow Zone" rather than merely reacting to immediate errors.

The framework establishes a robust closed-loop control system by fusing heterogeneous data:

- **Neuro-Cognitive Metrics:** Concentration level (att) is calculated via BCI spectral analysis: $att = \frac{\beta}{\alpha + \theta}$.
- **Kinetic Metrics:** Collaborative hit rates and synchronized reaction times.

The system recalibrates task parameters (Δd) every 20 seconds. If the team deviates from the Flow Zone (e.g., $att < att_{threshold}$), the DP-based controller re-optimizes the difficulty to prevent state transition to s_{break} .

3 SYSTEM IMPLEMENTATION AND EXPERIMENTAL VALIDATION

To evaluate the T-DDA framework, a multi-agent serious game was developed using the Unity3D engine. The game is designed as a controlled collaborative environment. The game features "Special Moles" that require synchronized strikes from both participants within a precise temporal window. This design operationalizes inter-member interdependency, a prerequisite for inducing group flow. The challenge level n is parameterized through the mole retraction Speed and the concurrent Quantity of targets. Higher levels necessitate faster reaction times and superior kinetic coordination between members.

The T-DDA controller recalibrates the task difficulty every 20-second observation window (T) by mapping team performance (p_T) and concentration (a_T) to the optimal difficulty path. Anxiety Mitigation: If $att < 60$ and $HitRate < 70\%$, the team is deemed "overwhelmed." The controller triggers a reduction in difficulty ($n \rightarrow n - 1$) to lower cognitive load. Flow Maintenance: If $70\% \leq HitRate \leq 80\%$, the team is within the "Flow Zone." The difficulty remains constant to maximize the duration of shared immersion. Boredom Intervention: If $HitRate > 80\%$ and concentration shows a downward trend, the system proactively elevates difficulty ($n \rightarrow n + 1$) to re-engage the participants.

Our user study ($N = 12$, 6 pairs, gender-balanced) was conducted to compare T-DDA against a static difficulty (Non-DDA) baseline. A within-subject design was employed with ABBA counterbalancing to eliminate order effects and learning biases. Post-session assessments utilized the Team Flow Model (TFM) questionnaire (7 dimensions, 30 items) and the Flow Short Scale (FSS) to capture multi-dimensional psychological data. Objective neuro-cognitive metrics were synchronized with subjective questionnaires to validate the framework's ability to bridge the "skill-challenge gap" in diverse teams.

4 RESULTS, DISCUSSION, AND CONCLUSION

The T-DDA framework demonstrated superior performance over static baselines across all group flow dimensions (Paired T-tests, $p < 0.05$):

- **Team Cohesion:** $t = 3.060, p < 0.05$. Adaptive balancing fostered shared agency.
- **Collective Ambition:** $t = 6.573, p < 0.001$. Proactive adjustment successfully prevented boredom.
- **Communication Efficiency:** $t = 4.660, p < 0.01$. Synchronized tasks required enhanced kinetic coordination.

Result: T-DDA reduced psychological distance from 6.33 to 5.69, acting as a social catalyst.

By solving the Bellman Equation via backward induction, T-DDA transcends reactive adjustment. The framework proactively steers the difficulty trajectory to keep the team within the "Flow Zone," effectively bridging the "coordination gap" caused by skill disparities. This research establishes a robust mathematical foundation for team-oriented difficulty adjustment. By integrating BCI-based neurofeedback and Dynamic Programming, we proved that optimized task challenges can directly enhance collective immersion and collaborative synergy in VR. Future efforts will focus on scalability for larger groups (3+) and the integration of Deep Reinforcement Learning to autonomously refine difficulty trajectories for complex, non-linear training scenarios.

5 ACKNOWLEDGMENTS

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